"Up and Running" With the Bridge2AI-Voice Dataset

Voice AI Symposium Hackathon, October 2026

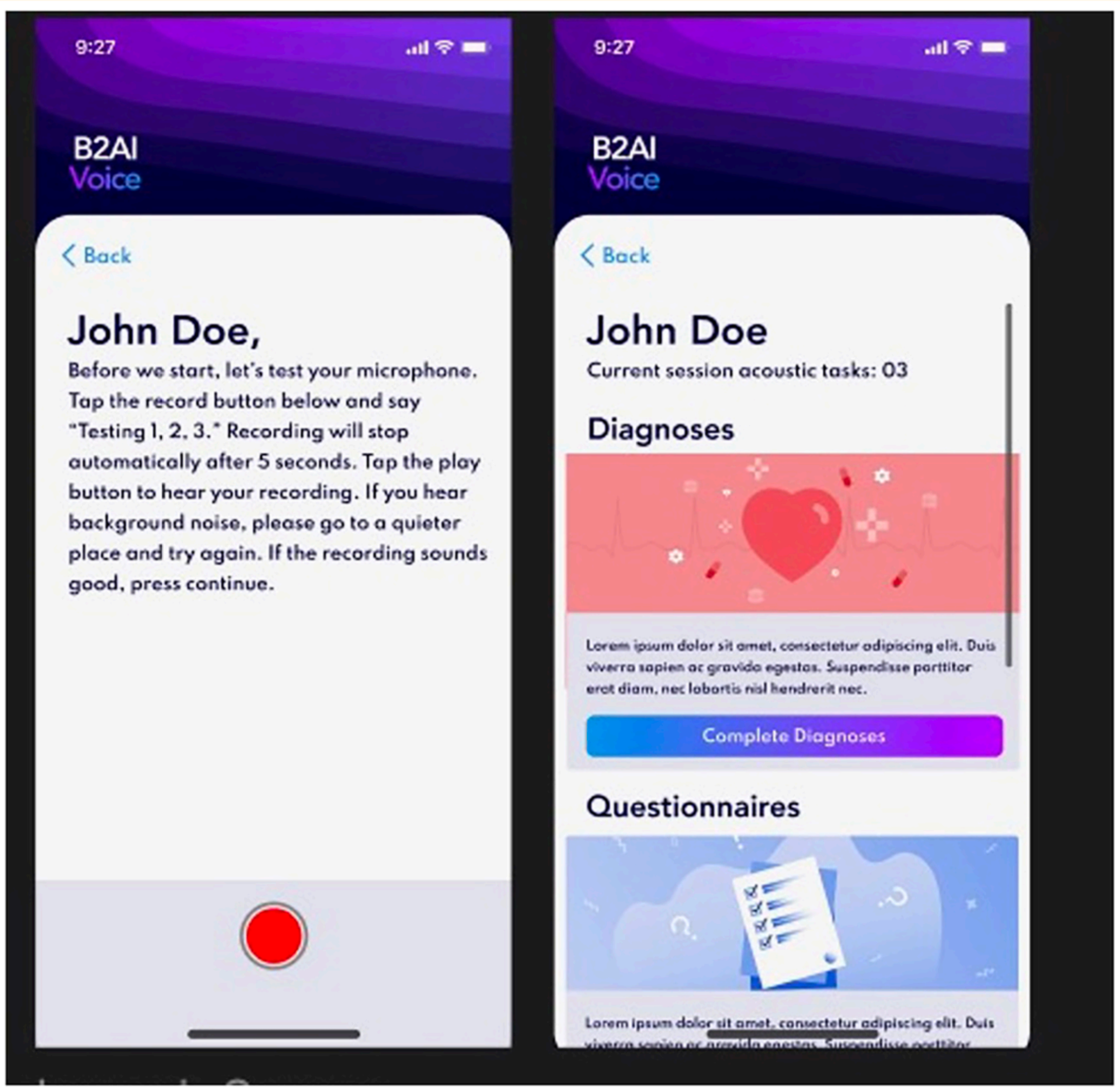
Steven Bedrick, PhD
Oregon Health & Science University

Today's Plan

Our dataset from 30,000 feet

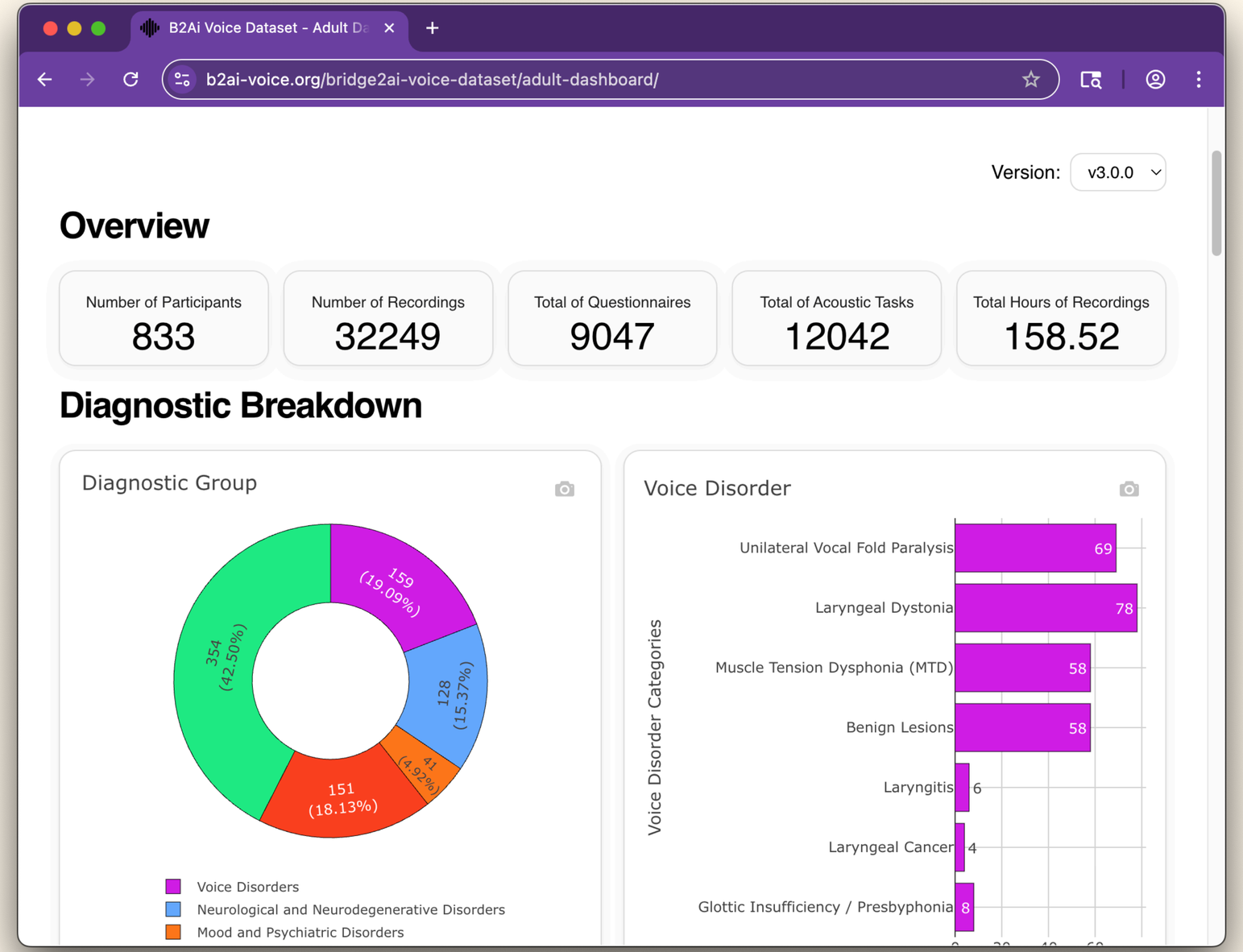
Getting hands dirty: problem formulation with our data

Practical demonstrations & exercises





Moothedan E, Boyer M, Watts S, Abdel-Aty Y, Ghosh S, Rameau A, et al. The Bridge2AI-voice application: initial feasibility study of voice data acquisition through mobile health. Front Digit Health. 2025 Apr 15;7:1514971.



https://docs.b2ai-voice.org/

The data are available in two different releases:

1. Derived features

   Credentialed access

   Questionnaire data, and extracted features from audio recordings for acoustic tasks

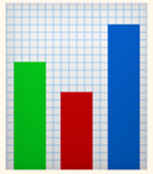
   ... including computed MFCCs and spectrograms.

2. Raw audio data

   Institutional DUA required

   Includes actual audio data from acoustic tasks

Both Releases:

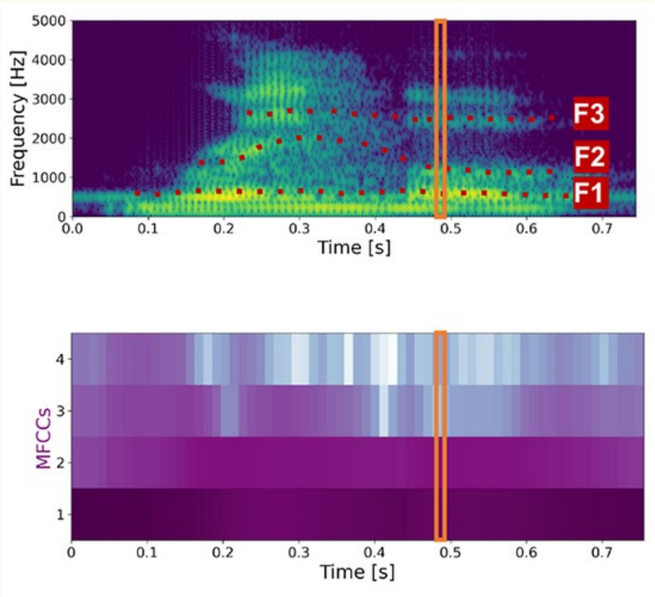






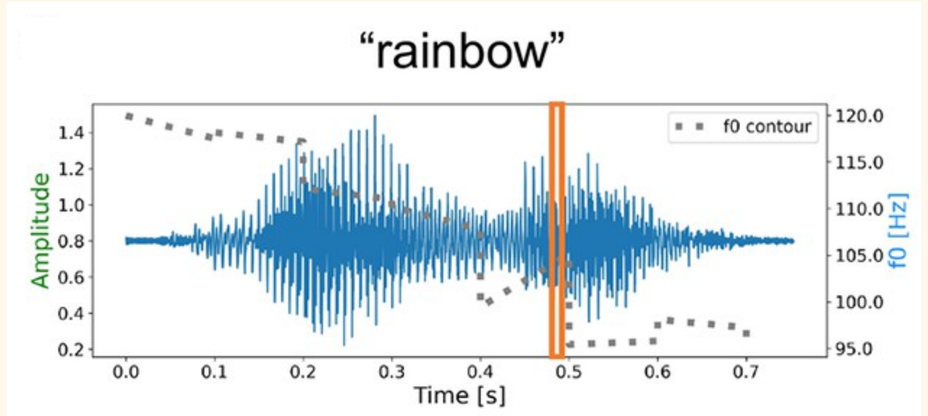
Questionnaire responses, dx status, demographics, confounders, etc.

1. Derived features



Computed stats: jitter, shimmer, spectral features, transcriptions, loudness, SI-SDR, etc.

2. Raw Audio



Low DM, Rao V, Randolph G, Song PC, Ghosh SS. Identifying bias in models that detect vocal fold paralysis from audio recordings using explainable machine learning and clinician ratings. PLOS Digit Health. 2024 May;3(5):e0000516.

B2AI VOICE

This repository is under review for potential modification in compliance with Administration directives.

Training opportunities for using the dataset: https://www.b2aivoicescholars.org/

Bridge2AI-Voice is a Precision Public Health grand challenge project funded by the NIH Common Fund Bridge2AI Program. Bridge2AI-Voice seeks to create a flagship, standardized, and ethically sourced dataset of 10,000 voices linked to health information to fuel research and discovery in voice biomarkers.

Our group aims to promote integration of voice as a biomarker of health in clinical care. To do so, we will generate a large multi-institutional, ethically sourced, and diverse voice dataset linked to multimodal health biomarkers to fuel voice AI research. Data collection is performed via a novel app (The Bridge2AI-Voice App) available as smartphone application linked to electronic health records (EHR). The app collected breathing sounds, voice, speech and linguistic tasks but also a considerable amount of health information through surveys and validated questionnaires. Other multimodal data collected includes imaging, genomics and respiratory function tests amongst others. The consortium is also addressing the growing ethical, legal, and social challenges surrounding voice AI, including risks of voice re-identification, vulnerabilities like voice AI hacking, concerns around voice data sharing and privacy, and the influence of gender and racial diversity on the development and application of these technologies.

Our developed best ethical practices through ethical inquiry have guided the development of voice collection protocol as well as data dissemination practices.

As voice is increasingly being recognized as a biomarker of health by the tech world and voice AI is gaining attention from multi-nationals such as Google, Amazon, Mozilla and Apple amongst others, many important issues related to patient privacy protection, ethical and fair representation of population, and clinical accuracy are arising. As a multidisciplinary group of academic experts, we aim to influence and guide the world of voice AI by ensuring patient protection through ethical and fairness principles and create safe innovative infrastructures to

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B2AI VOICE

Data Access

Access audio recordings

Instructions for accessing the data are provided on the data access page of the Bridge2AI learning module.

Derived dataset

A derived dataset containing spectrograms and combined phenotypic data is available on PhysioNet under a permission access mechanism. Registration on PhysioNet and signing of a data use agreement will enable access. The latest version of the dataset is available at the following URL:

Bridge2AI-Voice: An ethically-sourced, diverse voice dataset linked to health information

Dataset including raw audio

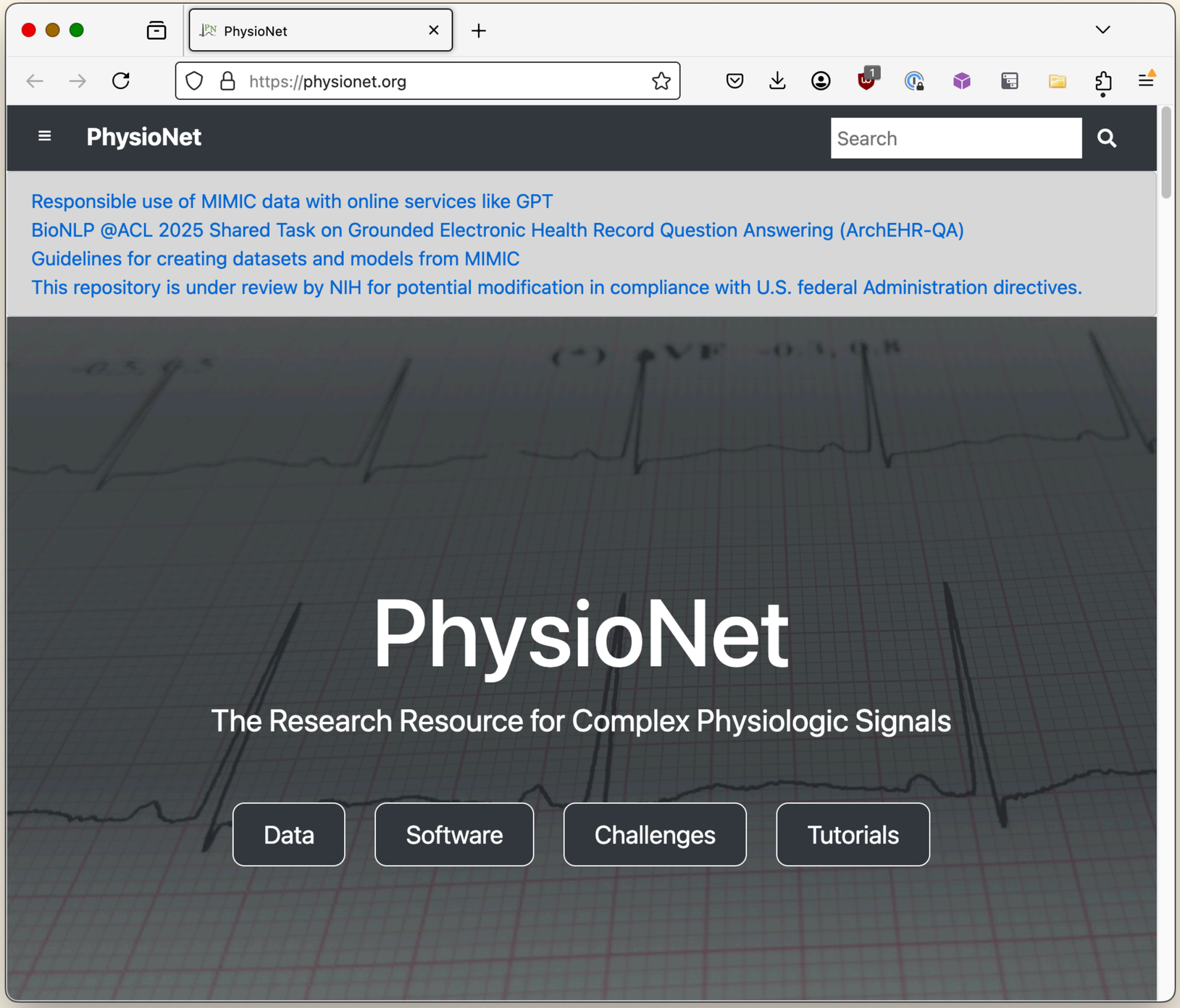
Raw audio is available under a controlled access mechanism. This process includes:

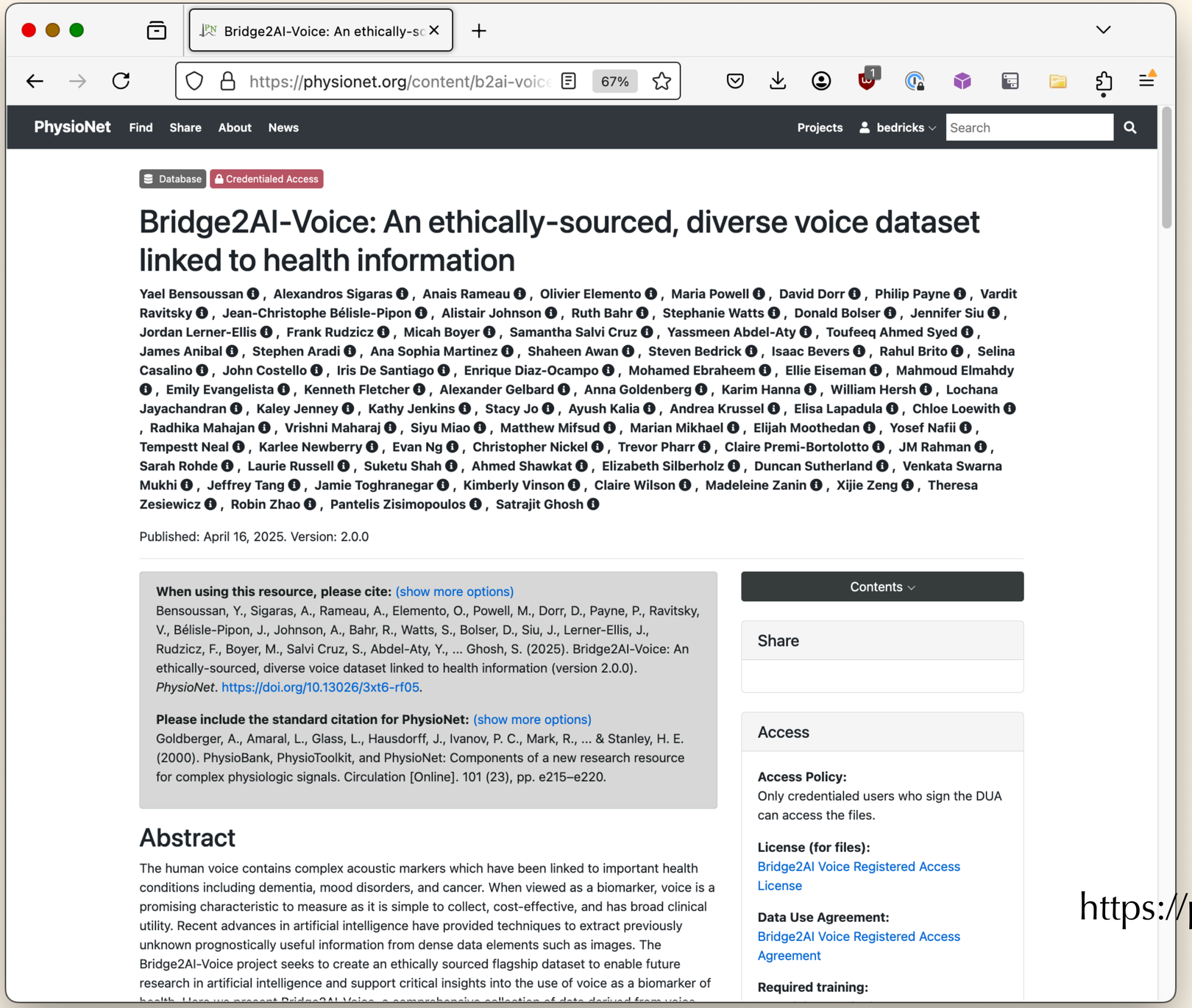
- Signing a Data Use Agreement (DUA): B2AI-Voice | Data Transfer and Use Agreement 2025.pdf
- Submitting a Data Access Request Form (DARF): B2AI-Voice | Data Access Request Form 2025.pdf

This process will require an institutional signature from a business official, a review by the Data Access Committee, and after approval, will be countersigned by USF. Once all forms are signed, the dataset can be obtained by submitting the counter-signed forms through PhysioNet at the following URL:

Bridge2AI-Voice: An ethically-sourced, diverse voice dataset linked to health information (Audio included)

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PhysioNet
https://physionet.org
PhysioNet
Search
Responsible use of MIMIC data with online services like GPT
BioNLP @ACL 2025 Shared Task on Grounded Electronic Health Record Question Answering (ArchEHR-QA)
Guidelines for creating datasets and models from MIMIC
This repository is under review by NIH for potential modification in compliance with U.S. federal Administration directives.
PhysioNet
The Research Resource for Complex Physiologic Signals
Data
Software
Challenges
Tutorials



https://physionet.org/content/b2ai-voice/

Bridge2AI Voice Registered Access Agreement

Authorized Researcher means the researcher that signs this data access agreement, and is responsible for ensuring that the Research Team respects the conditions of this agreement in using the Data.

Data means the data, in whole or in part, that is made accessible to You by the Data Provider on the basis of this Agreement, including metadata, and any enriched or derived data. It includes, but is not limited to, representations, observations, narratives or measurements that could be used as the basis for further analysis, calculation or reasoning, regardless of the manner of their expression or the medium used to represent them.

Data Provider means the legal entity, or organisation, that grants Authorized Researchers access to the Data, and acts as the formal data custodian responsible for performing oversight and stewardship functions relative to the Data.

Research Team means You and other qualified researchers **at Your institution** that collaborate in using the Data for research.

External Collaborators means qualified researchers **outside of Your institution** that collaborate in Your research, and who have completed a Registered Data Access Agreement and are duly authorized to access and use the Data.

1. **Purpose.**

You agree to use the Data exclusively for commercial and non-commercial research purposes. In completing this agreement, You confirm that You and your Research Team have sufficient training and research expertise to use the data in a manner that protects the rights and interests of research participants. You agree to remain accountable for your use of the Data, as well as the use thereof by members of Your Research Team.

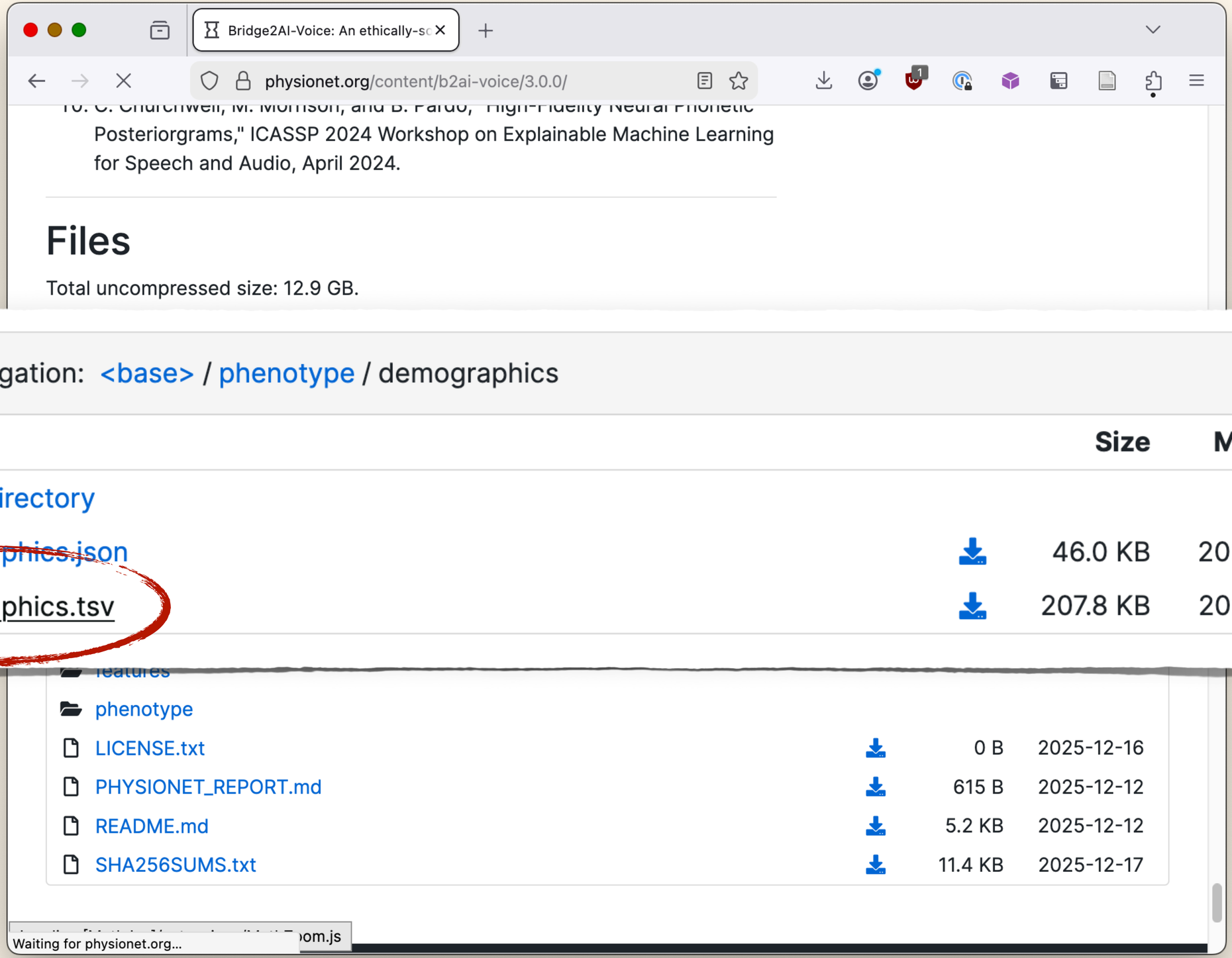
You agree to respect restrictions on data use arising from public law and private law, research ethics guidance, contracts, informed consent materials, consortium and institutional policies and research ethics requirements that are applicable to you.

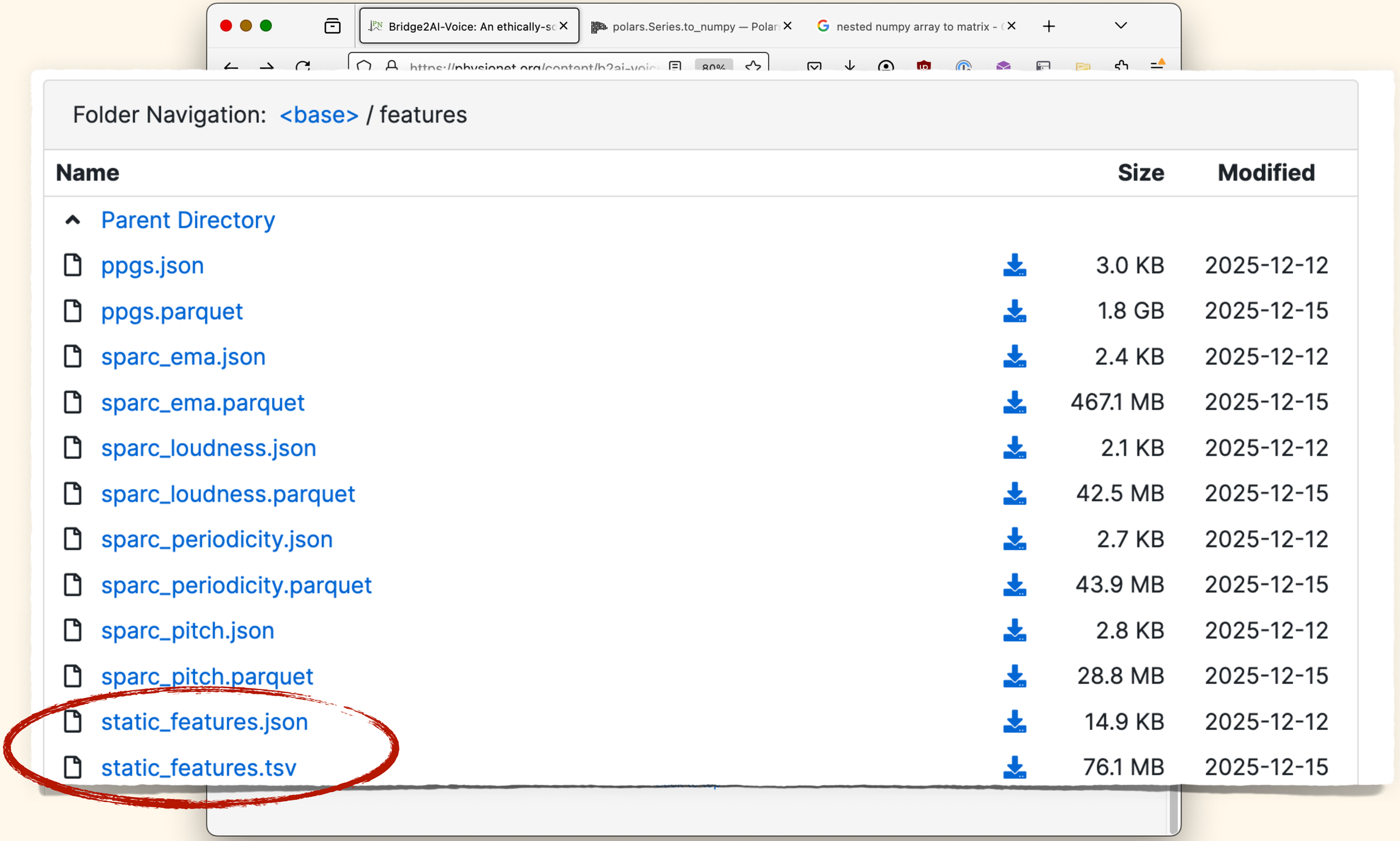
You agree to obtain all approvals and authorizations that the Data Provider requires you to hold, and maintain them throughout the entire period of data use and data retention.

2. **Scientific Publications and Research Outputs.**

In publishing the findings from Your research, You agree to include the attribution statement that is associated to the dataset. Such an attribution statement provides clear instructions on how to attribute the Research Team, and/or the database or biobank of origin, and/or the dataset.

In addition, You agree that the publications arising from Your research will not contain the personal data of research participants. Such publications will not contain information that could reasonably be expected to cause a research participant to experience harm or stigmatization.


Bridge2AI-Voice: An ethically-sc
physionet.org/content/b2ai-voice/3.0.0/
Posteriorgrams," ICASSP 2024 Workshop on Explainable Machine Learning for Speech and Audio, April 2024.
Files
Total uncompressed size: 12.9 GB.
Folder Navigation: <base> / phenotype / demographics
Name
Size
Modified
Parent Directory
demographics.json
46.0 KB
2025-12-15
demographics.tsv
207.8 KB
2025-12-17
phenotype
LICENSE.txt
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PHYSIONET_REPORT.md
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README.md
5.2 KB
2025-12-12
SHA256SUMS.txt
11.4 KB
2025-12-17
Waiting for physionet.org...


Bridge2AI-Voice: An ethically-s
polars.Series.to_numpy — Polar
nested numpy array to matrix -
80%
Folder Navigation: <base> / features
Name
Size
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Parent Directory
ppgs.json 3.0 KB 2025-12-12
ppgs.parquet 1.8 GB 2025-12-15
sparc_ema.json 2.4 KB 2025-12-12
sparc_ema.parquet 467.1 MB 2025-12-15
sparc_loudness.json 2.1 KB 2025-12-12
sparc_loudness.parquet 42.5 MB 2025-12-15
sparc_periodicity.json 2.7 KB 2025-12-12
sparc_periodicity.parquet 43.9 MB 2025-12-15
sparc_pitch.json 2.8 KB 2025-12-12
sparc_pitch.parquet 28.8 MB 2025-12-15
static_features.json 14.9 KB 2025-12-12
static_features.tsv 76.1 MB 2025-12-15

Important resources for working with this dataset:

Primary documentation site: https://docs.b2ai-voice.org/

Description of protocol, data use restrictions, study metadata, etc.

PhysioNet whitepaper: https://physionet.org/content/b2ai-voice/

Shorter and more focused overview, including brief code example.

RedCap repository: https://github.com/eipm/bridge2ai-redcap

Exhaustive description of all fields, examples of questionnaires, links to acoustic stimuli, map of protocols to questionnaires/acoustic tasks, etc.

FHIR schema: https://kind-lab.github.io/vbai-fhir/toc.html

Machine-readable metadata schema, for those of us who are into that sort of thing. 🤓

bridge2ai-redcap/README.m
github.com/eipm/bridge2ai-redcap/blob/main/README.md
New Chrome available



Voice as a Biomarker of Health

bridge2ai-redcap

REDCap Data Dictionary and Metadata for the Bridge2AI project.



Version	Date YYYY-mm-dd	Data Dictionary	Metadata	PDF's (English)	PDF's (Spanish)
v4.3.0	2025-10-24	Data Dictionary	Metadata	Files	Files

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📚 How to Cite

> Bensoussan, Y., Ghosh, S. S., Rameau, A., Boyer, M., Bahr, R., Watts, S., Rudzicz, F., Bolser, D., Lerner-Ellis, J., Awan, S., Powell, M. E., Belisle-Pipon, J.-C., Ravitsky, V., Johnson, A., Zisimopoulos, P., Tang, J., Sigaras, A., Elemento, O., Dorr, D., ... Bridge2AI-Voice. (2024). eipm/bridge2ai-redcap. Zenodo. https://zenodo.org/doi/10.5281/zenodo.12760724

Task Overview

#	Task Name	Type	Estimated Time to Complete (mm:ss) ⏳	Protocol 📓	Total # of Recordings 🔊 (if Acoustic Task)	Questionnaire Form PDF Link / Acoustic Task Description Youtube Link
1	Demographics	Questionnaire 📕	03:00	Generic Protocol	-	PDF
2	Confounders	Questionnaire 📕	12:00	Generic Protocol	-	PDF
3	Voice Perception	Questionnaire 📕	00:30	Generic Protocol	-	PDF
4	VHI-10	Questionnaire 📕	01:00	Generic Protocol	-	PDF
5	Patient Health Questionnaire-9	Questionnaire 📕	01:30	Generic Protocol	-	PDF
6	GAD-7 Anxiety	Questionnaire 📕	01:30	Generic Protocol	-	PDF
7	Prolonged Vowel	Acoustic Task 🎤	01:00	Generic Protocol	1	YouTube Link
8	Maximum Phonation Time	Acoustic Task 🎤	02:00	Generic Protocol	3	YouTube Link

Q Generic Confounders

Record ID

Questionnaire - Metadata

Session ID

Questionnaire Started At

Questionnaire Completed At

Questionnaire Duration (seconds)

Smoking

Have you smoked at least 100 cigarettes in your entire life? (There are 20 cigarettes in a pack.)?
- ◯ Yes
- ◯ No
- ◯ Don't know
- ◯ Prefer not to answer

Do you now smoke cigarettes every day, some days, or not at all?
- ◯ Every day
- ◯ Some days
- ◯ Not at all
- ◯ Don't know
- ◯ Prefer not to answer

How old were you when you first started regular cigarette smoking?
- ◯ Specify
- ◯ Don't know
- ◯ Prefer not to answer

Age:

If you have completely stopped smoking cigarettes, about how old were you when you stopped?
- ◯ Specify
- ◯ Don't know
- ◯ Prefer not to answer

Age when you stopped smoking:

How many years have you or did you smoke cigarettes?
- ◯ Specify
- ◯ Don't know
- ◯ Prefer not to answer

Number of years:

Native Hawaiian or other Pacific Islander
- ☐ Polynesian
- ☐ Micronesian
- ☐ Melanesian
- ☐ Other Pacific Islander

White
- ☐ European
- ☐ Middle Eastern Or North African
- ☐ Arab

Canadian Indigenous or Aboriginal
- ☐ First Nation
- ☐ Inuk/inuit
- ☐ Metis

If race not listed above, please specify:

What is your ethnic or cultural origin?
- ◯ Hispanic or Latino
- ◯ Not Hispanic or Latino
- ◯ Prefer not to answer

Check all that apply
- ☐ Spaniard
- ☐ Mexican
- ☐ Central American
- ☐ South American
- ☐ Latin American
- ☐ Puerto Rican
- ☐ Cuban
- ☐ Dominican

Education

What is your highest level of education?
- ◯ No formal education
- ◯ Some elementary school
- ◯ Some secondary or high school education
- ◯ High School or secondary school degree complete
- ◯ Some college education
- ◯ Associate's or technical degree complete
- ◯ College or baccalaureate degree complete
- ◯ Some post-baccalaureate education
- ◯ Graduate or professional degree complete
- ◯ Doctoral or post graduate education
- ◯ Other
- ◯ Prefer not to answer

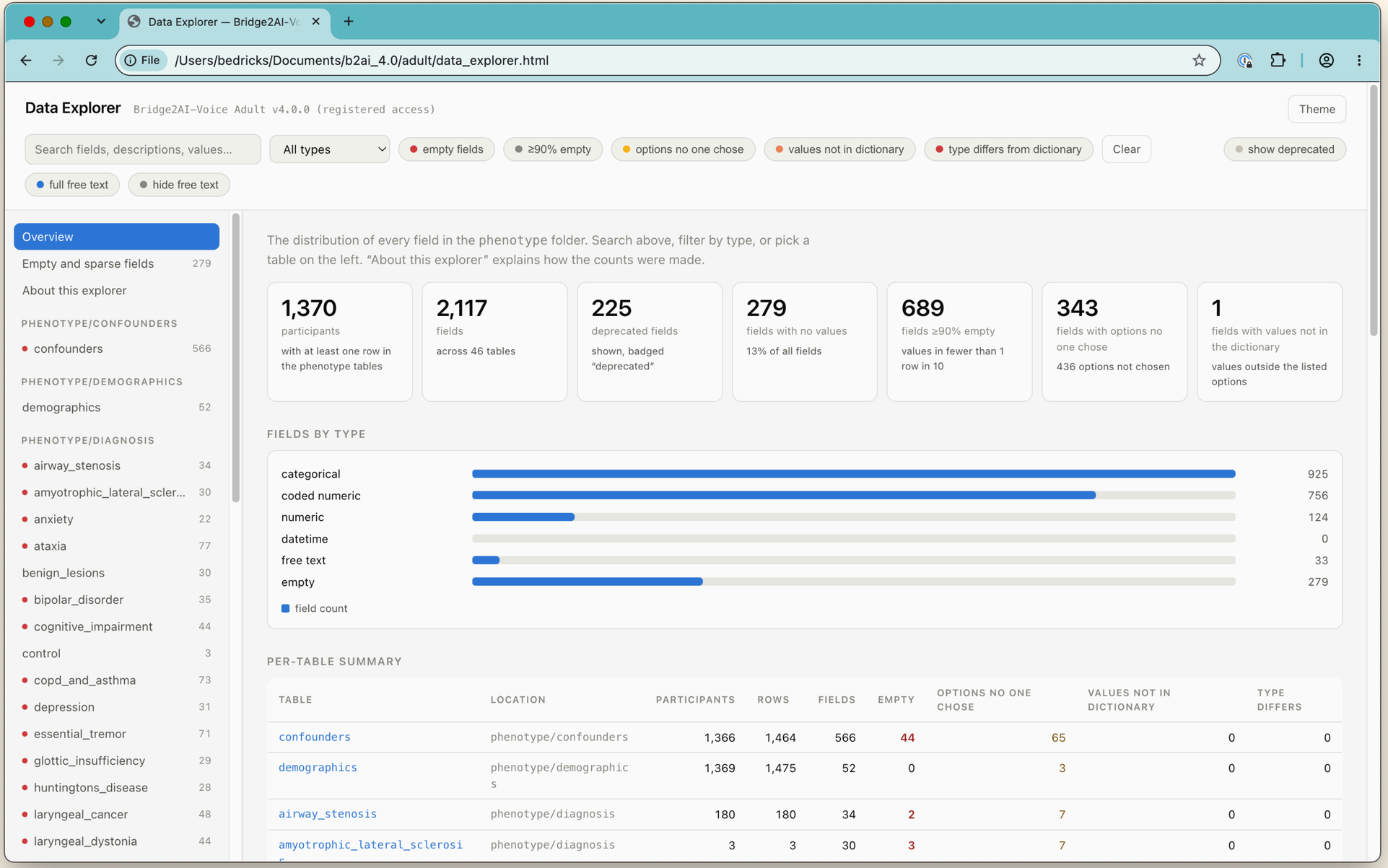
If you selected "other" level of education, please specify:

Disability Questions

Are you deaf or do you have serious difficulty hearing?
- ◯ No
- ◯ Yes
- ◯ Prefer not to answer

Are you blind or do you have serious difficulty seeing, even when wearing glasses?
- ◯ No
- ◯ Yes
- ◯ Prefer not to answer

The dataset itself comes with tools:



Today's Plan

Our dataset from 30,000 feet

Getting hands dirty: problem formulation with our data

Practical demonstrations & exercises

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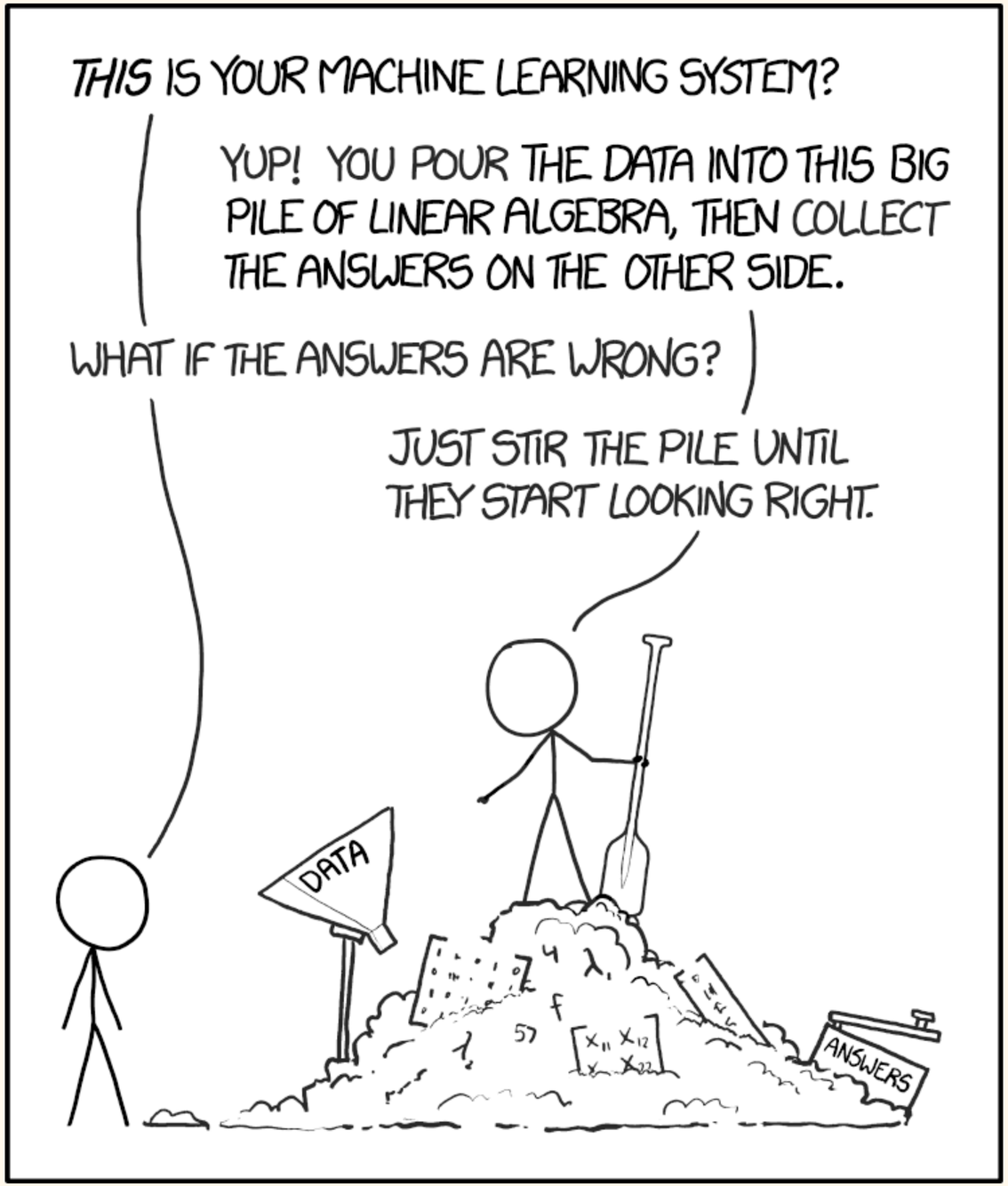
Getting hands dirty: problem formulation with our data

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General workflow for using Bridge2AI-Voice Dataset:

1. Blindly load entire dataset into computer
2. Pick a disorder column, train classifier on all other features
3. Make ROC curve
4. Obtain VC funding
5. Get investigated by the FDA, FTC, and SEC
6. Upon being pardoned, retire to Barbados

... and they all lived happily ever after...



https://xkcd.com/1838/

General workflow for using Bridge2AI-Voice Dataset:

1. Identify comparison of interest

   Disorder vs. control? Disorder vs. disorder? Disorder vs. age?

2. Identify appropriate protocols and stimuli

   Voice disorder protocol? Neuro protocol? Generic protocol?

   Alternatively: start with stimulus, find relevant protocols

General workflow for using Bridge2AI-Voice Dataset:

3. Develop participant cohort

Inclusion criteria to consider:

Eligibility for protocol of interest

Case/control status

Sex at birth

Clinical confounders and covariates

4. EDA to verify sample size, data completeness for stimuli, etc.

General workflow for using Bridge2AI-Voice Dataset:

5. Identify and collect patient and task-level features

   Phenotypic (age, sex at birth, smoking status, etc.)

   Task-specific acoustic features, MFCC/spectrogram, etc.

   This might involve reading published literature, etc.

"NOW, MAY WE SHOVE ALL THESE FEATURES INTO OUR ML WOODCHIPPER?"

6. Do whatever you were hoping to do vis-a-vis classification, regression, clustering, etc.

Let's walk through an example!

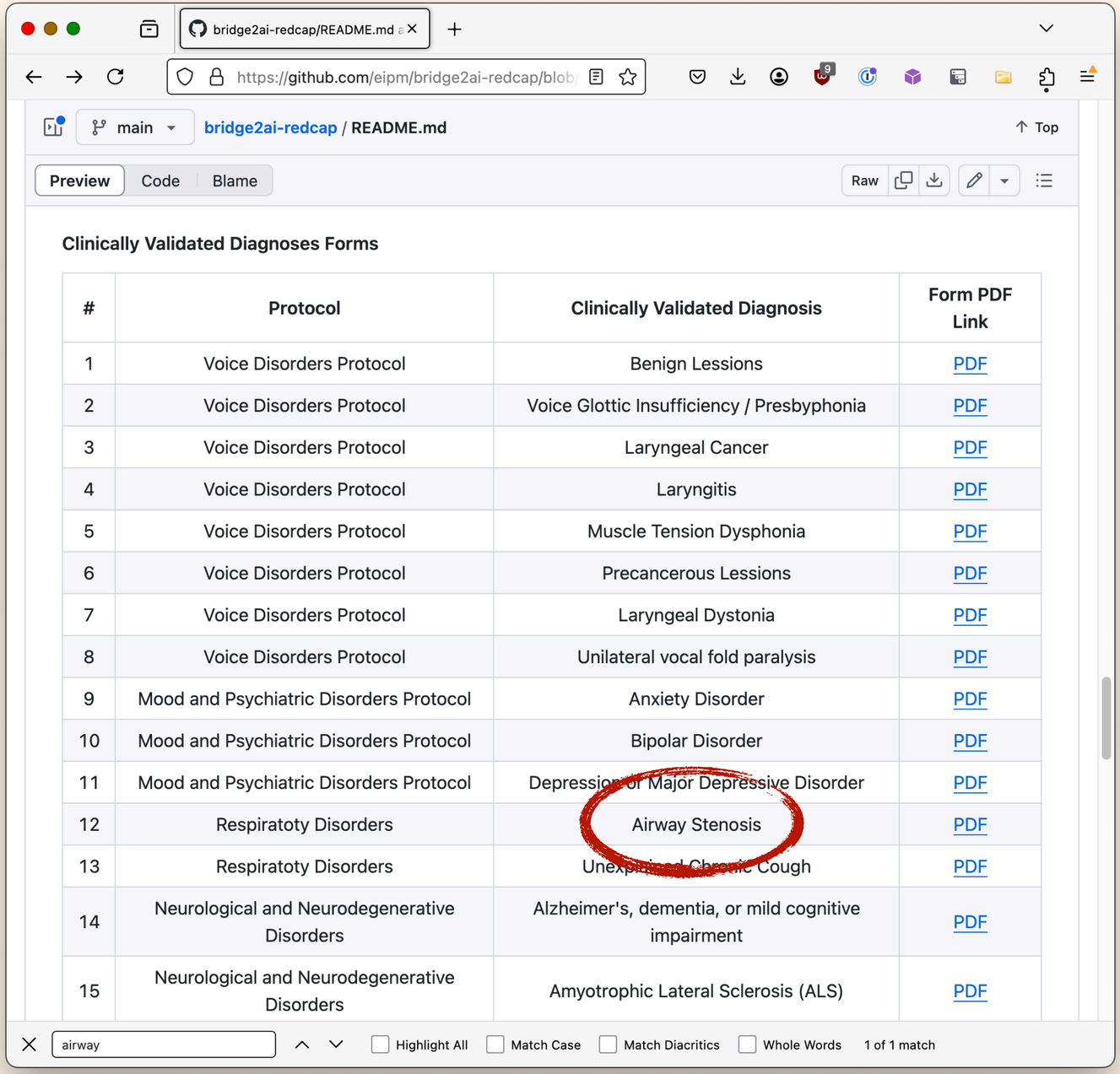
RQ: How do jitter and shimmer differ in patients with airway stenosis, as compared to controls?

1. Comparison of interest:

    Airway stenosis vs. control population

2. Determine protocols and stimuli:

    Consult the documentation...



Clinically Validated Diagnoses Forms

#	Protocol	Clinically Validated Diagnosis	Form PDF Link
1	Voice Disorders Protocol	Benign Lessions	PDF
2	Voice Disorders Protocol	Voice Glottic Insufficiency / Presbyphonia	PDF
3	Voice Disorders Protocol	Laryngeal Cancer	PDF
4	Voice Disorders Protocol	Laryngitis	PDF
5	Voice Disorders Protocol	Muscle Tension Dysphonia	PDF
6	Voice Disorders Protocol	Precancerous Lessions	PDF
7	Voice Disorders Protocol	Laryngeal Dystonia	PDF
8	Voice Disorders Protocol	Unilateral vocal fold paralysis	PDF
9	Mood and Psychiatric Disorders Protocol	Anxiety Disorder	PDF
10	Mood and Psychiatric Disorders Protocol	Bipolar Disorder	PDF
11	Mood and Psychiatric Disorders Protocol	Depression or Major Depressive Disorder	PDF
12	Respiratoty Disorders	Airway Stenosis	PDF
13	Respiratoty Disorders	Unexplained Chronic Cough	PDF
14	Neurological and Neurodegenerative Disorders	Alzheimer's, dementia, or mild cognitive impairment	PDF
15	Neurological and Neurodegenerative Disorders	Amyotrophic Lateral Sclerosis (ALS)	PDF

Confidential

D Resp Airway Stenosis

Record ID _____

1. Gold Standard Diagnosis

Has the patient had pulmonary function testing within the past 3 months?
- ◯ No
- ◯ Yes
- ◯ Not certain

Please upload the PFT document

Does the patient have airway stenosis confirmed by a laryngoscopy, bronchoscopy or CT imaging?
- ◯ No
- ◯ Yes
- ◯ Not certain

Confirmed by
- ☐ Confirmed by bronchoscopy or laryngoscopy
- ☐ Confirmed by CT imaging

2. Disease Subtype

Disease Subtype
- ◯ Subglottic Stenosis
- ◯ Bilateral Vocal fold immobility or Glottic Stenosis
- ◯ Supraglottic Stenosis
- ◯ Tracheal Stenosis
- ◯ Multi-Level Upper Airway Stenosis

Bilateral Vocal fold immobility or Glottic Stenosis Subtype
- ◯ Posterior Glottic Stenosis
- ◯ Bilateral Vocal Fold Paralysis
- ◯ Anterior glottic Web
- ◯ Not certain or TBD
- ◯ Other

Other, Please specify _____

3. Etiology

What is the etiology?
- ◯ Not certain or TBD
- ◯ Iatrogenic (surgery or intubation)
- ◯ Idiopathic
- ◯ Auto-immune or granulomatous disease
- ◯ Radiation
- ◯ GERD
- ◯ Caustic Injury or Inhalation Injury
- ◯ Infectious (Diphtheria, TB, Bacterial Tracheitis, other)
- ◯ Other

Other, Please specify _____

Let's walk through an example!

2. Determine protocols and stimuli:

Airway stenosis cases, + controls from resp. protocol

Part A speech tasks, plus some breath/cough.

3. Inclusion criteria...

Let's hop over to R...

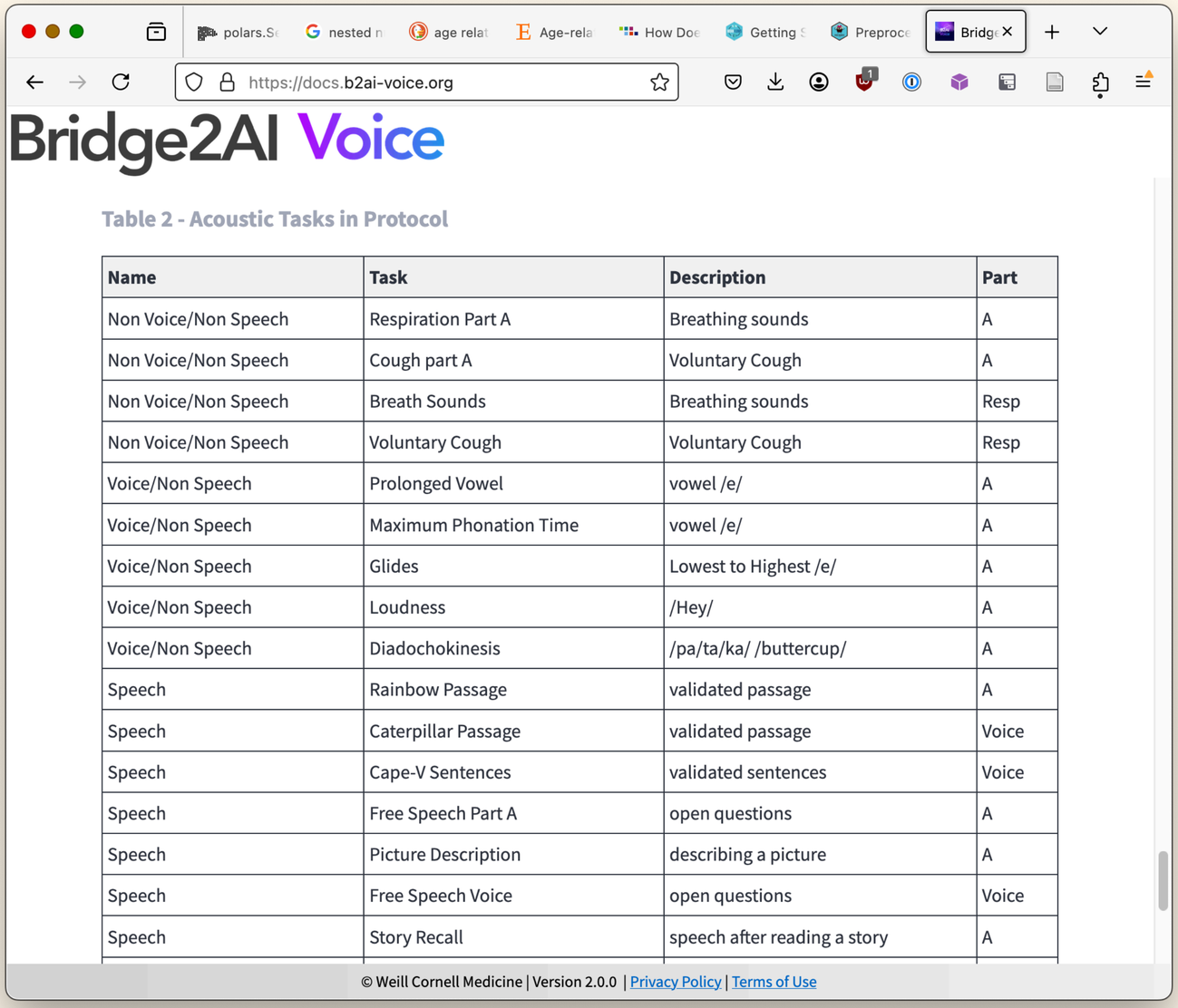


Table 2 - Acoustic Tasks in Protocol

Name	Task	Description	Part
Non Voice/Non Speech	Respiration Part A	Breathing sounds	A
Non Voice/Non Speech	Cough part A	Voluntary Cough	A
Non Voice/Non Speech	Breath Sounds	Breathing sounds	Resp
Non Voice/Non Speech	Voluntary Cough	Voluntary Cough	Resp
Voice/Non Speech	Prolonged Vowel	vowel /e/	A
Voice/Non Speech	Maximum Phonation Time	vowel /e/	A
Voice/Non Speech	Glides	Lowest to Highest /e/	A
Voice/Non Speech	Loudness	/Hey/	A
Voice/Non Speech	Diadochokinesis	/pa/ta/ka/ /buttercup/	A
Speech	Rainbow Passage	validated passage	A
Speech	Caterpillar Passage	validated passage	Voice
Speech	Cape-V Sentences	validated sentences	Voice
Speech	Free Speech Part A	open questions	A
Speech	Picture Description	describing a picture	A
Speech	Free Speech Voice	open questions	Voice
Speech	Story Recall	speech after reading a story	A

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https://dmice.ohsu.edu/bedricks/b2ai_f2f_oct_2026/

Thank you!

Questions?

bedricks@ohsu.edu

A huge "thank you" to:

The *entire* Bridge2AI-Voice team, across all of our sites

Our participants, for donating their data and time

The rest of the Bridge2AI Consortium

The NIH! We love you guys! 3OT2OD032720

